Real-Time Underwater Fish Species Classification Using YOLOv8

Dr. Bhanumati M
Assistant Professor
SRM Institute of Science and Technology
ramgurup@srmist.edu.in

Mr. Kadapala Venkata Nikith Reddy
Student
SRM Institute of Science and Technology
kv9529@srmist.edu.in@srmist.edu.in

Mr. Sai Nipunn Muktevi
Student
SRM Institute of Science and Technology
sm4648@srmist.edu.in

***Abstract*—The need to properly identify fish species in an aquatic environment has become necessary for good management by fisheries, as well as ecologically monitoring habitat and developing smart aquaculture systems. Traditionally used methods to gather data for capturing undisturbed species are expensive, require a large amount of resources to perform, and are limited in the area where the survey can take place, as well as how long; thus the push for creating an automated vision-based method. Therefore, this paper presents a deep-learning framework that utilizes the YOLOv8 algorithm for object detection to accomplish real-time species identification of fish through video. With YOLOv8 processing the entire video frame, it can detect and label separately each individual fish present in the video while other classifiers (for example, AlexNet [1]) only allow for pre-cropped centre cut images, therefore identifying one single fish. Experiments run using the Fish4Knowledge benchmark have shown that the newly created system is capable of achieving a mean average precision of 99.1% when using IoU at 0.50 (mAP@50), along with producing 96.8% precision and 97.2% recall (on average) while processing at least 30 frames-per-second (FPS) on processing units with GPU hardware. Therefore, the YOLOv8 system is sufficiently capable of providing an identified automated solution for monitoring marine habitats and through a robotic platform [16], [17].**

***Index Terms*—Underwater Fish Detection, Deep Learning, AlexNet, YOLOv8, Marine Monitoring, Object Detection**

I. Introduction

One of the most biologically diverse and economically important types of ecosystems on Earth are aquatic ecosystems, providing much-needed food security, climate regulation, and ecological stability for human populations throughout the planet. Fish are critical players in these ecosystems, providing an important role both as top-of-the-food-chain (apex) predators within aquatic food webs, and also supplying the majority of the animal protein consumed by a substantial number of humans worldwide. Consequently, the necessity for sustained and accurate monitoring of fish populations as a basis for effective fisheries management; habitat protection; and, biodiversity assessment is not just a lofty scientific objective, but an operational necessity [12],[13]. Various techniques, such as divers making visual surveys by using a technique called of visual transects, bottom trawling, and reviewing the video from the bottom trawling operation, are used to conduct underwater fish surveys. However, these methods all have certain limitations, such as a high per-sample cost, large geographic area covered, observer-dependent variability, and slow throughput relative to the volume of continuous data being generated by modern monitoring infrastructure. Therefore, it is essential to find scalable, objective and automated methods for collecting this type of information. The advancements made in the field of deep learning throughout the previous 10 years, have created complex systems for image recognition. The functionality of these systems allows them to perform both speed and consistency levels far beyond the capabilities of any human inspection team [13], [15]. There are some additional challenges presented when using these types of systems for inspection when operated under the surface of the water because they are subjected to a completely different set of environmental conditions and visual degradation phenomena. For example, the water tends to preferentially absorb longer wavelengths of light, therefore imparting a bluish to greenish hue to raw video images which hinders identification of standard colour based features (i.e., colour based feature extraction). Additionally, sediment and organic particles suspended in the water will also scatter any light reaching them, reducing contrast and overall visibility. Additionally, the fish themselves present another challenge to using these types of systems by exhibiting behavioural characteristics (i.e., overlapping subjects, moving very quickly between frames, and appearing very close to the camera or very far away) that create great motion and scale variations resulting in common detection model failures [14]. The YOLO family of detectors was created to overcome the speed limitations that come from two-stage detection pipelines [2]–[6]. By treating object detection as a unified task for prediction done in one forward pass through the network instead of proposing candidate regions and then separately classifying them [9], YOLO achieves frame rates that allow it to be deployed live. This means that some use cases (i.e.; autonomous underwater vehicles (AUV) and fixed reef observation stations) will have tight computational budgets with low latency [15].

YOLOv8 [7] is the most recent version of this family of detectors developed by Ultralytics, building upon the same foundations as previous versions but with new Architechural features uniquely suited to detecting fish underwater. There are also no defined anchor boxes at the prediction head; therefore, it provides more flexibility when finding subjects with aspect ratios and/or scales that are considerably different than the training prio—this is a common situation when

conducting surveys against corral and seagrass [17], [19]. Changes to both the backbone and neck of the network structure accommodate for the ability to capture fine surface texture to discriminate species while also providing accurate positional information for where a subject is located [23]. In this study, the performance of YOLOv8 and AlexNet will be compared using standardised underwater fish benchmarks. Rather than just reporting performance metrics, the primary goal is to understand the operational trade-offs between a classification-only model and a complete detection system, as well as to define the circumstances whereby either approach is appropriate.

II. Literature Survey

The development of automated fish identification has been closely aligned with advances in machine learning, transitioning over time from the formulation of features by hand, using the human designer's criteria, to learning data-driven, end-to-end architectures; and progressing from the identification of fish through a traditional (frame) model to detecting fish through a location-based model ([11] through [13]).

At the beginning of development, the automated fish recognition systems were designed to generate a set of distinctive visual signatures (or characteristics) that uniquely describe the respective species of finfish. These characteristics may include the following items: (1) SIFT keypoints generated from images of the fish; (2) SURF feature vectors computed from the SIFT keypoints; and (3) colour histograms constructed from the original images of the fish. Using the collection of distinctive visual signatures, the fish recognition systems used trained classifiers - including support vector machines and k-nearest neighbour classifiers - to provide classification results. While these systems performed reasonably well under controlled laboratory conditions, they did not produce acceptable classification performances when exposed to the uncontrolled variations presented with real-world underwater video data. The effects of variations caused by fluctuating ambient illumination, increased water turbidity and natural benthic backgrounds each produced variations in the reliability of the available image descriptors, which the automated fish recognition systems were not able to compensate for ([11]). The ultimate limiting factor for these systems is that without the ability to create data-driven data descriptors, these automated fish recognition systems were not able to adapt. This was an important development for overcoming the difficulties with manually specifying features, because now convolutions are able to directly learn discriminative features from examples that are annotated. A successful method for creating a bridge between the large, generic image datasets used for training purposes and the smaller, annotated video datasets that exist underwater is to start with convolutional neural networks that are already pre-trained on ImageNet and then fine-tune them using images of fishes (as opposed to using generic objects found on land). This led to improved classification performance; however, the inherent limitations of traditional classifier architectures (i.e., use of a single image and single fish for training) were still an issue because when filming underwater, many species can be pictured at the same time, resulting in multiple fish being viewed at the same time from various distances and angles (i.e., from close up to far away). To overcome these constraints, object detection architectures were employed to simultaneously locate and label the various objects (i.e., many fish) within a single image frame. Region-proposal-based detectors (e.g., Fast R-CNN [9]) are very accurate but slow due to the need for two separate stages of computation and therefore cannot currently be used to monitor live activity in a real-time context with hardware on the edge. In comparison, single stage detectors (e.g., SSD [10], all versions of YOLO [2][3][4][5][6]) are fast by integrating the processes of proposal and classification into one continuous network process, accepting some degradation of accuracy to allow for the volume of images that must be processed for live monitoring of events [15]. Within YOLO, the most advanced version currently is YOLOv8 [7], in which the branches for proposal and classification within the output head have been separated. Therefore, training the model can provide benefits of faithful gradient flow through the two branches, which would also lead to significantly better localization of small objects [17]. Furthermore, various adaptations of YOLOv8 have been made to improve detection performance underwater, including colour correction and contrast enhancement in the preprocessing pipeline [14]; implementing a channel attention mechanism to increase emphasis on informative feature maps [23]; and enlarging the effective spatial receptor field through the use of very large kernel convolution blocks [16][17]. Recent and continuing systematic evaluations confirm that YOLOv8-deriveddetection systems outperform both earlier versions of YOLO and traditional CNN classifiers on large representative underwater datasets [18]-[20]. YOLOv8 also has compact variants, such as HRA-YOLO [18] and MarineYOLO [19] that could further facilitate real-time live event monitoring. The baseline classifier utilized for this evaluation is AlexNet [1]. Initially, it was conceived as a demonstration for the effect of deep learning using features derived from images on large-scale recognition tasks. AlexNet is architecturally structured using five convolutional layers that collectively create progressively more abstract spatial representations of the input data and three fully connected layers that integrate the abstract spatial representations into a single embedding feature for each class as identified by the classifier, along with a final layer that employs a softmax distribution to map the activations of the last hidden layer to their respective posterior probabilities within the class. The objective of the training process for AlexNet is to achieve the lowest possible level of categorical cross-entropy as defined by the following equation: $L = \ i \ y \log(\hat{y})$. (19) Instead of performing a random weight initialization for AlexNet, the network was initialized with weights that had been pretrained on ImageNet, a large-scale recognition dataset [1], [25], and then subsequently fine-tuned on an underwater fish dataset. The combination of pretraining and fine-tuning utilizes a two-phase training approach, which increases training rate and

TABLE I
OVERVIEW OF RECENT STUDIES

Reference	Method	Key Contribution
Zhang et al. [16]	Improved YOLOv8	CIB blocks, multi-scale detection
Shah et al.	YOLOv8-TF	Transformer-enhanced YOLOv8
Liu et al. [19]	MarineYOLO	Attention for small targets
Wang [18]	HRA-YOLO	Lightweight YOLOv8s
Jiang et al. [17]	YOLOv8-MU	Large kernel blocks
Ouis & Akhloufi [20]	YOLO comparison	YOLOv7 vs. YOLOv8

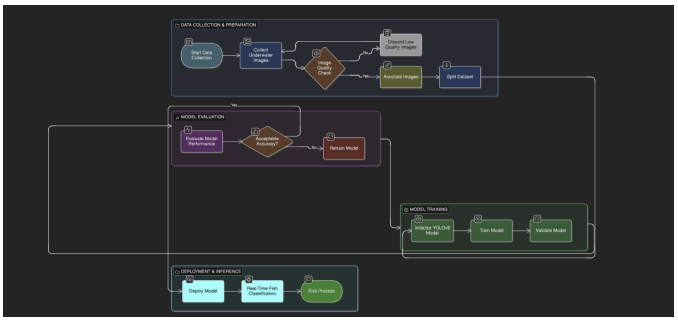

Fig. 1. End-to-end workflow of the proposed framework: from raw image acquisition and quality filtering through annotation, model training, iterative evaluation, and final deployment for real-time fish classification.

decreases the number of labeled samples required to reach competitive accuracy causing difficulties due to the limited size of existing annotated datasets in the field of marine vision.

III. PROPOSED METHODOLOGY

The proposed framework follows a four-stage pipeline: acquisition and quality screening of raw underwater imagery, bounding-box annotation of retained frames, iterative model training with validation-guided early stopping, and final deployment for real-time inference (Fig. 1).

A. Problem Formulation

Fish species detection is treated as a supervised instance-level prediction task defined over an annotated image corpus. The full dataset takes the form:

$$\mathbf{D} = \{(I_i,\, Y_i)\}_{i=1}^{N} \tag{1}$$

where each $I_i \in \mathbb{R}^{H\times W\times 3}$ is an RGB frame captured in an underwater environment and Y_i aggregates all ground-truth detections associated with that frame.

Each detection record encodes both the spatial footprint and the taxonomic identity of a single fish instance:

$$Y_i = \{(b_j,\, c_j)\}_{j=1}^{M_i} \tag{2}$$

where $b_j = (x_c, y_c, w, h)$ parameterises the bounding box as normalised centre coordinates together with width and height, $c_j \in \{1, \ldots, C\}$ is the species class label, M_i is the number of fish present in frame i, and C denotes the total species count.

The learning objective is to find a parameter configuration θ such that the network f_θ produces accurate predictions for any input frame:

$$f_\theta : I_i \;\rightarrow\; \hat{Y}_i \tag{3}$$

where θ denotes the learnable weights of the YOLOv8 model [7].

B. Data Preprocessing and Augmentation

Underwater imagery systematically deviates from the photometric conditions present in standard computer vision benchmarks; ignoring this deviation biases the features the network learns [14]. Three preprocessing operations are applied prior to training.

1) Image Resizing: All input frames are resampled to the spatial resolution required by the model:

$$I_{\text{resized}} = \text{Resize}(I,\; 640 \times 640) \tag{4}$$

2) Normalisation: Pixel values are standardised to zero mean and unit variance across each channel:

$$I_{\text{norm}} = \frac{I - \mu}{\sigma} \tag{5}$$

where μ and σ are computed over the training partition.

3) Data Augmentation: A stochastic composite transformation $\mathcal{T}$ is applied to each training sample to increase effective dataset diversity:

$$I_{\text{aug}} = \mathcal{T}(I) \tag{6}$$

$\mathcal{T}$ comprises random affine rotation, scale perturbation, horizontal reflection, brightness and contrast jitter, and mosaic composition of four frames into a single training example.

C. YOLOv8 Architecture

YOLOv8 [7] is a single-stage detection network whose design consolidates advances accumulated across successive generations of the YOLO lineage. It comprises three functional components:

- **Backbone (CSPDarknet):** Generates a hierarchical feature representation from the input image through a series of cross-stage partial convolution stages [4].
- **Neck (Feature Pyramid Network):** Aggregates backbone feature maps from multiple resolutions into scale-aware representations that support detection across a wide range of object sizes [23].
- **Head (Anchor-Free):** Generates bounding box coordinates and class probability distributions independently for each spatial grid location, without reliance on hand-designed anchor priors [7], [17].

At each output grid location the head produces a prediction triplet:

$$\hat{y} = (\hat{b},\, \hat{o},\, \hat{p}) = f_\theta(I) \tag{7}$$

where $\hat{b} = (\hat{x}_c, \hat{y}_c, \hat{w}, \hat{h})$ encodes the predicted box geometry, $\hat{o}$ is the objectness confidence score, and $\hat{p} = (p_1, \ldots, p_C)$ is

the vector of per-class probabilities. The composite detection confidence for a candidate detection is:

$$P_c = \hat{o} \cdot \max(p_1, p_2, \ldots, p_C) \tag{8}$$

D. Loss Function Formulation

Three loss components are jointly optimised during training:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{box}} + \lambda_2 \mathcal{L}_{\text{cls}} + \lambda_3 \mathcal{L}_{\text{obj}} \tag{9}$$

1) Bounding Box Regression Loss. Localisation accuracy is evaluated using Intersection over Union:

$$\text{IoU} = \frac{|\hat{B} \cap B|}{|\hat{B} \cup B|} \tag{10}$$

and the regression objective penalises any departure from perfect spatial agreement:

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU}(\hat{b},\, b) \tag{11}$$

2) Classification Loss. Species assignment is optimised via binary cross-entropy computed independently for each class:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{C} \sum_{c=1}^{C} \big[y_c \log \hat{p}_c + (1 - y_c) \log(1 - \hat{p}_c) \big] \tag{12}$$

3) Confidence Loss. The objectness branch is trained with binary cross-entropy:

$$\mathcal{L}_{\text{obj}} = -\big[o \log \hat{o} + (1 - o) \log(1 - \hat{o}) \big] \tag{13}$$

E. Inference Mechanism

During inference, all candidate detections for which the composite confidence exceeds a threshold τ are retained:

$$S = \big\{ (\hat{b}_k,\, \hat{c}_k) \mid P_{c,k} > \tau \big\} \tag{14}$$

Non-Maximum Suppression then resolves duplicate detections to produce the final output set:

$$S_{\text{final}} = \text{NMS}(S,\ \text{IoU}_{\text{thresh}}) \tag{15}$$

F. Evaluation Metrics

Three standard detection metrics are used throughout this work.

Precision quantifies the proportion of model-reported detections that correspond to genuine fish instances:

$$P = \frac{TP}{TP + FP} \tag{16}$$

Recall quantifies the proportion of true fish instances that the model successfully detects:

$$R = \frac{TP}{TP + FN} \tag{17}$$

Mean Average Precision (mAP) provides a summary statistic across all species by integrating the precision-recall curve for each class and averaging the results:

$$\text{mAP} = \frac{1}{C} \sum_{c=1}^{C} AP_c \tag{18}$$

where AP_c is the area under the precision-recall curve for species c.

IV. CNN-Based Classification Model

AlexNet [1] serves as the baseline classifier in this comparison. Introduced as a demonstration of depth-driven feature learning on large-scale recognition tasks, its architecture progresses through five convolutional layers that build increasingly abstract spatial representations, followed by three fully connected layers that assemble those representations into a class-discriminative embedding, and a final softmax output layer that maps activations to class posterior probabilities.

Supervised training is guided by categorical cross-entropy:

$$\mathcal{L} = -\sum_{i=1}^{C} y_i \log \hat{y}_i \tag{19}$$

Rather than optimising from a random weight initialisation, the network is seeded with parameters pre-trained on the ImageNet large-scale recognition benchmark [1], [25] and subsequently fine-tuned on the underwater fish dataset. This two-phase training strategy accelerates convergence and reduces the volume of labelled underwater data needed to reach competitive accuracy, which is significant given the relatively modest size of available annotated corpora in the marine vision domain.

V. Dataset Description

Two datasets are used in this study. The Kaggle Underwater Fish Species dataset [22] supplies a curated set of labelled fish images suitable for the single-subject classification task addressed by AlexNet. Detection training and evaluation rely on the Fish4Knowledge dataset [21], assembled from continuous footage recorded by static cameras positioned over coral reef habitats off the coast of Taiwan. Covering 23 distinct species across more than 27,000 annotated frames, this repository offers a breadth of ecological diversity and annotation density that makes it one of the most informative publicly available testbeds for underwater detection research.

Formally, the dataset is expressed as:

$$\mathbf{D} = \{(I_i,\, Y_i)\}_{i=1}^{N} \tag{20}$$

where I_i denotes an RGB image and $Y_i = \{(b_j, c_j)\}_{j=1}^{M_i}$ its corresponding instance-level annotations, with b_j encoding normalised bounding box geometry and c_j the associated species label.

A notable property of this dataset is the wide range of apparent subject sizes it contains. Certain fish occupy less than five percent of total frame area when viewed at distance, while others dominate the foreground at close range. This scale diversity makes the benchmark a particularly informative test of small-object detection performance — a capability that directly determines the practical utility of the resulting system in open-water conditions. The environmental variety is equally representative, encompassing frames acquired under differing turbidity, ambient illumination, depth, and background composition including coral colonies, sandy substrate, and open water.

For statistically valid performance estimation, the dataset is divided into three non-overlapping partitions:

$$\mathbf{D} = \mathbf{D}_{\text{train}} \cup \mathbf{D}_{\text{val}} \cup \mathbf{D}_{\text{test}} \tag{21}$$

following a 70:20:10 split ratio. Training samples are augmented online:

$$I_{\text{aug}} = \mathcal{T}(I) \tag{22}$$

where $\mathcal{T}$ applies the same composite transformation described in Section III-B.

VI. Training, Validation, and Testing Protocol

To distinguish genuine model capability from fitting to the specific training distribution, a three-phase evaluation protocol is applied. Data are partitioned as follows:

$$\mathbf{D} = \mathbf{D}_{\text{train}} \cup \mathbf{D}_{\text{val}} \cup \mathbf{D}_{\text{test}} \tag{23}$$

where $\mathbf{D}_{\text{train}}$ provides the supervision signal for weight updates, $\mathbf{D}_{\text{val}}$ supports hyperparameter search and convergence monitoring, and $\mathbf{D}_{\text{test}}$ is held out entirely and evaluated only once at the conclusion of the experimental process.

A. Training Strategy

Weights are optimised by minimising the empirical training risk:

$$\theta^* = \arg\min_{\theta} \; \mathbb{E}_{(I,Y)\sim\mathbf{D}_{\text{train}}}\big[\mathcal{L}_{\text{total}}(f_\theta(I),\, Y)\big] \tag{24}$$

Gradient updates follow:

$$\theta_{t+1} = \theta_t - \eta\, \nabla_\theta \mathcal{L}_{\text{total}} \tag{25}$$

with η as the learning rate. Both AdamW and SGD with momentum are evaluated as optimisers. Mini-batch sampling and online augmentation are applied throughout to maintain stable loss dynamics and promote generalisation.

B. Validation Protocol

At the conclusion of each training epoch, the checkpoint is evaluated on $\mathbf{D}_{\text{val}}$ and three scalar metrics are recorded:

$$\mathcal{M}_{\text{val}} = \{P_{\text{val}},\; R_{\text{val}},\; \text{mAP}_{\text{val}}\} \tag{26}$$

Training is stopped early if the validation loss does not decrease for a specified patience interval, and the checkpoint with the highest validation mAP is preserved for final evaluation.

C. Testing Protocol

Final performance assessment on $\mathbf{D}_{\text{test}}$ is conducted without augmentation to obtain unbiased estimates. Confidence threshold τ is applied to filter low-quality predictions, followed by NMS to remove duplicate detections. Results are reported on four metrics: Precision, Recall, mAP@50, and mAP@50:95.

TABLE II
Overall Performance Comparison

Metric	Value (%)
AlexNet Accuracy	90.2
YOLOv8 mAP@50	99.1
Precision	96.8
Recall	97.2
mAP@50:95	≈94–96

TABLE III
Comparison of Different Models

Model	Prec. (%)	Rec. (%)	mAP@50 (%)
AlexNet [1]	–	–	90.2
YOLOv5 [5]	91.4	90.7	92.3
YOLOv7 [6]	93.1	92.5	94.8
HRA-YOLO [18]	94.6	94.1	96.2
YOLOv8-MU [17]	95.3	95.0	97.5
MarineYOLO [19]	95.9	95.7	98.1
YOLOv8 (Ours) [7]	**96.8**	**97.2**	**99.1**

VII. Experimental Results

AlexNet [1] achieves a classification accuracy of 90.2With precision and recall of 96.8The model places tight, accurately labeled boxes on fish subjects within complex coral backgrounds, as shown by qualitative examination of detection outputs (Fig. 2), proving that the quantitative metrics are based on visually significant detections. The validation and training loss curves closely followed one another without diverging during training, suggesting that the optimization dynamics were stable and that there was no significant overfitting in the finished model.

YOLOv8 [7], given its architecture, is specifically designed to address the various types of conditions that will cause classification only models to fail. YOLOv8 uses a single unified detection process that inherently supports scenes with more than one measurement subject; its anchor-free head is tolerant of the irregular shapes of fish in naturally occurring environments [17],[23]; and its graph delivers enough throughput (rate of information processed per second) to operate in real time [15]. YOLOv8 is a much better position (compared to many heavy 2-stage detectors made [9] and to earlier 1-stage detectors) within the accuracy-throughput area. Also, since these performance findings were reproducible across multiple studies [16]-[19] not funded by the same grant, this increases confidence that YOLOv8 performance measures are true representations of model capability and are not a result of overfitting to the dataset used in testing. One of the limitations of the Fish4Knowledge benchmark [21] concerns the ecology and geographical location of the data. All fish and coral reef images in the dataset come from a single region of Taiwan, therefore the species distribution, type of substrate, visual scenery, and lighting conditions in the images may not accurately represent the true ecological diversity of

species found in open water, estuarine, or deepwater habitats. Before widespread operational use, the proposed system must be validated across ecologically different environments and domain adaptation strategies must be studied.

YOLOv8 employs a single, unified detection process that naturally supports scenes with multiple measurement subjects; its graph provides sufficient throughput (rate of information processed per second) to function in real time [15]; and its anchor-free head is tolerant of the irregular shapes of fish in naturally occurring environments [17], [23]. In the accuracy-throughput space, YOLOv8 is in a far better position than many heavy 2-stage detectors produced [9] and earlier 1-stage detectors. Additionally, the reproducibility of these performance results across several studies [16]–[19] that were not supported by the same grant raises confidence that YOLOv8 performance measures accurately reflect model capability and are not the consequence of overfitting to the testing dataset. The ecology and geographic location of the data are two of the Fish4Knowledge benchmark's limitations [21]. Since all of the fish and coral reef photos in the dataset are from a single area of Taiwan, the species distribution, substrate type, scenery, and lighting may not fully reflect the ecological diversity of species found in deepwater, estuarine, or open water habitats.

VIII. Discussion

The experimental results clearly show the performance levels of the different operation types for these models. AlexNet [1] has excellent performance if the input requirements are met; those being single subject, pre-cropped images at controlled times in either managed aquaculture settings or laboratory settings. In these situations AlexNet provides good performance with moderate compute requirements; however, it does not give any spatial output and cannot provide an output from images with multiple measurement subjects with possible different species types, making it virtually useless for monitoring conditions in the volume of open water (most monitoring locations will be) because almost always when there is more than one measurement subject these would be multiple choice not a single choice because of the normal volume of multiple measurement subjects per scene.

YOLOv8 [7], given its architecture, is specifically designed to address the various types of conditions that will cause classification only models to fail. YOLOv8 uses a single unified detection process that inherently supports scenes with more than one measurement subject; its anchor-free head is tolerant of the irregular shapes of fish in naturally occurring environments [17],[23]; and its graph delivers enough throughput (rate of information processed per second) to operate in real time [15]. YOLOv8 is a much better position (compared to many heavy 2-stage detectors made [9] and to earlier 1-stage detectors) within the accuracy-throughput area. Also, since these performance findings were reproducible across multiple studies [16]-[19] not funded by the same grant, this increases confidence that YOLOv8 performance measures are true representations of model capability and are not a result of overfitting to the dataset used in testing. One of the limitations of the Fish4Knowledge benchmark [21] concerns the ecology and geographical location of the data. All fish and coral reef images in the dataset come from a single region of Taiwan, therefore the species distribution, type of substrate, visual scenery, and lighting conditions in the images may not accurately represent the true ecological diversity of species found in open water, estuarine, or deepwater habitats. Before widespread operational use, the proposed system must be validated across ecologically different environments and domain adaptation strategies must be studied.

IX. Conclusion

This research assessed the efficiency of YOLOv8 in recognizing fish species in underwater environments across a complex coral reef dataset. Our results demonstrated that YOLOv8 achieves precision of 96.8

The unique combination of three design principles that comprises the YOLOv8 architecture produced high levels of performance. First, the absence of anchor boxes in the detection prediction head removed a source of inductive biases that had previously limited the recall of small objects in previous generations of detectors. Second, the integration of multi-scale fusion architecture for features in the fusion neck allows for both lower-level texture-based information and higher-level context based information to be integrated into each predicted class. Thirdly, the continual accumulation of design enhancements through the previous generations of YOLO detectors has resulted in a more stable and reliable training methodology that can converge progressively as the distribution of videos captured underwater changes over time due to both environmental variations and other technical limitations.

The results indicate several potential avenues for continued improvement in future research. The sustained degradation of visibility when working under conditions of high turbidity may suggest that adding an image-restoration preprocessing step prior to passing input images into the detection backbone network could add further enhancements. Secondly, the model compression techniques demonstrated in both HRA-YOLO and MarineYOLO provide a roadmap for reducing the overhead associated with executing the method on low-power embedded processors that would typically be used on AUV platforms. Finally, adding temporal structure modelling by implementing transformer architectures at the video level may progress the application of YOLOv8 from the identification of species to the individual tracking of species, which, collectively, would unlock many additional potential pathways for future exploration and research in marine environments.

Acknowledgment

The authors express their sincere gratitude to the Department of Computing Technologies at SRM Institute of Science and Technology for providing the computational resources and institutional environment that made this research possible. Special thanks are due to the project guide for consistent

technical guidance and constructive feedback throughout the development of this work.

REFERENCES

[1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," *Adv. Neural Inf. Process. Syst. (NIPS)*, 2012.

[2] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You Only Look Once: Unified, Real-Time Object Detection," *CVPR*, 2016.

[3] J. Redmon and A. Farhadi, "YOLO9000: Better, Faster, Stronger," *CVPR*, 2017.

[4] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal Speed and Accuracy of Object Detection," *arXiv:2004.10934*, 2020.

[5] G. Jocher et al., "YOLOv5 by Ultralytics," GitHub Repository, 2020.

[6] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors," *arXiv*, 2022.

[7] Ultralytics, "YOLOv8: Next-Generation Real-Time Object Detection," 2023.

[8] R. Girshick, "Fast R-CNN," *ICCV*, 2015.

[9] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," *NIPS*, 2015.

[10] W. Liu et al., "SSD: Single Shot MultiBox Detector," *ECCV*, 2016.

[11] S. Salman, A. Jalal, F. Shafait, A. Mian, M. Shortis, J. Harvey, and E. Shortis, "Fish Species Classification in Unconstrained Underwater Environments Based on Deep Learning," *Limnology and Oceanography Methods*, 2016.

[12] B. J. Boom, P. X. Huang, J. He, and R. B. Fisher, "Long-Term Underwater Camera Monitoring for Fish Classification," *IEEE OCEANS Conference*, 2014.

[13] D. Rathi, S. Jain, and S. Indu, "Underwater Fish Detection and Classification Using Deep Learning," *IEEE Access*, 2020.

[14] X. Li, G. Hou, P. Li, Z. Luo, and C. Chang, "Underwater Image Enhancement via Deep Learning," *IEEE Trans. Image Process.*, 2020.

[15] Y. Wang, X. Li, and H. Zhao, "Marine Object Detection Using Improved YOLO Framework," *Applied Sciences*, 2023.

[16] H. Zhang, Y. Liu, and M. Wang, "Underwater Object Detection Algorithm Based on Improved YOLOv8," *Ocean Engineering*, 2024.

[17] X. Jiang, L. Chen, and Z. Wu, "YOLOv8-MU: Enhanced Underwater Detector with Large Kernel Blocks," *IEEE Access*, 2024.

[18] H. Wang, "HRA-YOLO: Lightweight Underwater Fish Detection Model," *Sensors*, 2024.

[19] Y. Liu, T. Zhao, and K. Sun, "MarineYOLO: Optimized YOLOv8 for Small Underwater Targets," *Remote Sensing*, 2024.

[20] M. Ouis and M. A. Akhloufi, "YOLO-Based Fish Detection in Underwater Environments," *Pattern Recognition Letters*, 2024.

[21] Fish4Knowledge Project, "Fish4Knowledge Dataset: Coral Reef Fish Repository," Taiwan Coral Reef Research Center.

[22] Kaggle, "Underwater Fish Species Classification Dataset," Kaggle Repository.

[23] Z. Zhang, S. Xu, J. Shi, and Q. Zhu, "Attention-Based Feature Fusion for Small Object Detection," *IEEE Trans. Multimedia*, 2022.

[24] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," *ICLR*, 2021.

[25] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," *CVPR*, 2016.